

6. Refraction of Light

(Total marks: 7)

Refraction of Light

- **Introduction:** Light generally travels in a straight line. When an opaque object blocks its path, a shadow is formed. However, light can bend under certain conditions, leading to the phenomenon known as refraction.

Try This: (Materials: Glass, 5 Rupee Coin, Pencil, Metallic Vessel, etc.)

Activity 1:

1. Fill a transparent glass with water.
2. Dip a portion of a pencil vertically into the water and observe the thickness of the submerged portion.
3. Now, incline the pencil to the water surface and observe its thickness.

- **Observation:**

- The submerged portion of the pencil appears thicker than the portion above water.
- When inclined, the pencil appears broken near the water's surface.

[Refraction of light in water - YouTube](#)

Explanation: These effects are due to the change in the direction of light when it comes out of the water. This bending of light is known as the refraction of light.

Activity 2:

1. Place a 5 rupee coin in a metallic vessel.
2. Slowly move away from the vessel until the coin disappears from view.
3. Keep looking in the direction of the coin.
4. Ask a friend to slowly fill the vessel with water.
 - **Observation:** You will be able to see the coin again once the water level reaches a certain height.

[ADD WATER SEE COIN - ENGLISH - Refraction of light! - YouTube](#)

Explanation: This phenomenon is also due to the refraction of light.

Activity 3:

1. Place a glass slab on a blank piece of paper and draw its outline (PQRS).
2. Draw an inclined straight line on the side of PQ, intersecting it at N. Place two pins vertically at points A and B along this line.
3. Look at the pins from the opposite side of the slab and place two more pins (C and D) vertically so that the images of A and B are in line with C and D.

4. Remove the glass slab and the pins, then draw a straight line through points C and D that intersects SR at M.
5. Join points M and N. Observe the incident ray AN and emergent ray MD.

[Refraction of light through a glass slab - YouTube](#)

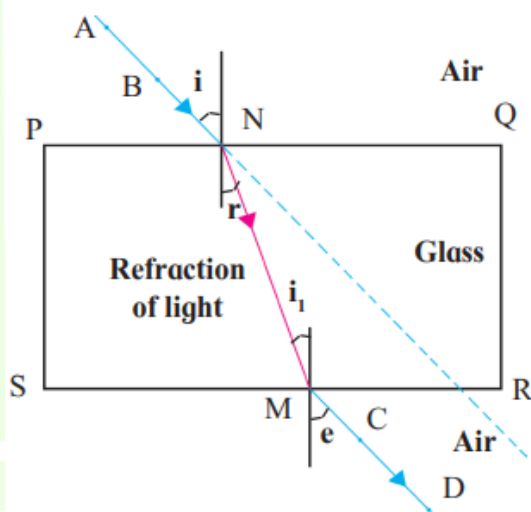
Refraction of Light Through a Glass Slab

Key Concepts:

- **Refraction of Light:** The bending of light as it passes from one medium to another.

Refraction Process:

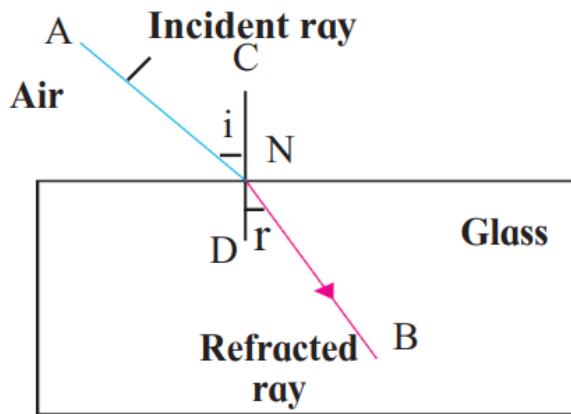
1. **First Refraction:** When light enters the glass from air at point N on the side PQ, the angle of incidence is i , and the angle of refraction is r .
2. **Second Refraction:** When light exits the glass into the air at point M on the side SR, the angle of refraction at N is i_1 and equals r .



- **Key Observation:**
 - $i_1 = r, e = i$
 - The light ray MD, emerging from the glass slab, is parallel to the incident ray AN but displaced from it.

Laws of Refraction

1. **Incident Ray and Refracted Ray:** At the point of incidence (N), these rays are on opposite sides of the normal to the surface of the slab (CD). The incident ray, refracted ray, and the normal lie in the same plane.



2. Snell's Law:

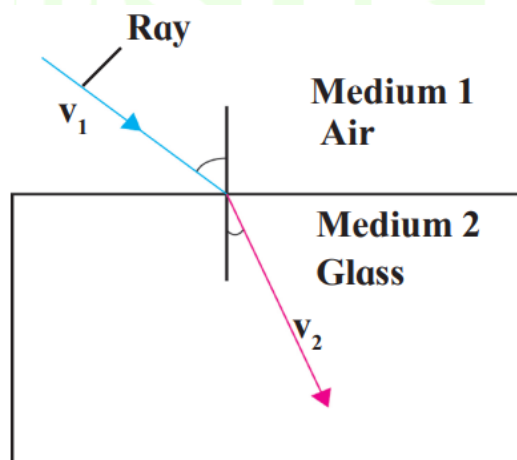
$$\frac{\sin i}{\sin r} = \text{constant} = n$$

- $\rightarrow$ ratio of $\sin(i)$ to the $\sin(r)$ is constant.
- n is the refractive index of the **second medium with respect to the first**.
- For a ray incident along the normal ($i=0$), the light goes straight without bending ($r=0$).

Refractive Index

- **Definition:** The ratio of the speed of light in a medium to the speed of light in the another medium.
- **Dependence:** Refractive index depends on the velocity of light in the medium. The value of the refractive index varies for different media and for different colors of light within the same medium.
- **Formula:**
- The refractive index of the **second medium with respect to the first medium**

$$\text{Refractive index } {}_2n_1 = \frac{\text{Velocity of light in medium 1 } (v_1)}{\text{Velocity of light in medium 2 } (v_2)}$$



6.3 Light ray going from medium 1 to medium 2

$${}_1n_2 = \frac{v_2}{v_1}$$

Similarly, the refractive index of medium 1 with respect to medium 2 is

Absolute Refractive Indices of Some Media

Substance	Refractive Index	Substance	Refractive Index	Substance	Refractive Index
Air	1.0003	Fused Quartz	1.46	Carbon disulphide	1.63
Ice	1.31	Turpentine oil	1.47	Dense flint glass	1.66
Water	1.33	Benzene	1.50	Ruby	1.76
Alcohol	1.36	Crown glass	1.52	Sapphire	1.76
Kerosene	1.39	Rock salt	1.54	Diamond	2.42

Refractive Index

- **Absolute Refractive Index:**

- If the first medium is a vacuum, then the refractive index of medium 2 is called the absolute refractive index and is denoted as n .

Refraction of light in different media:

- **When a light ray passes from a rarer medium to a denser medium:**
 - It bends **towards** the normal.
- **When a light ray passes from a denser medium to a rarer medium:**
 - It bends **away** from the normal.
- **When a light ray is incident normally at the boundary between two media:**
 - It does not change its direction and hence does not get refracted.

Mirage and Atmospheric Refraction (Fig 6.5)

- **Mirage:**
 - A mirage is an optical illusion caused by the refraction of light in the atmosphere.
 - **Cause:** The air near the hot road or desert surface is heated, making it less dense and causing the refractive index to decrease near the ground. As light rays pass through varying layers of air (hot and cold), they bend, creating the illusion of water or a distorted image of distant objects.

[what is the mirage? - YouTube](#)

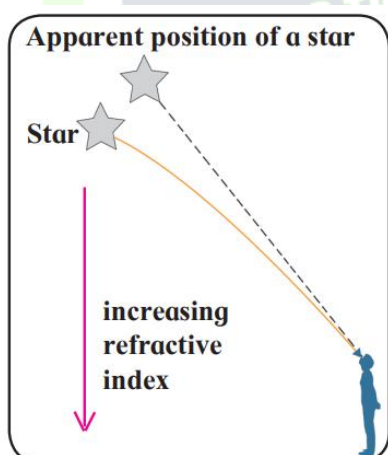
Effect of Atmospheric Conditions on Twinkling of Stars

- **Apparent Position of a Star (Fig 6.6):**

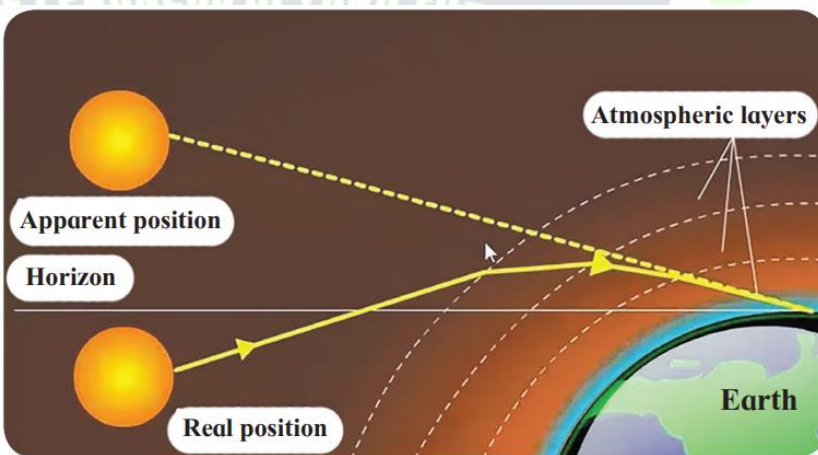
- Stars appear to be higher in the sky than their actual position due to the refraction of light in the Earth's atmosphere.
- **Cause:** As light from a star passes through the Earth's atmosphere, it moves from a rarer medium to a denser one, bending towards the normal. This continuous bending makes the star appear to be in a higher position than it actually is.

- **Effect of Atmospheric Refraction on Twinkling:**

- The apparent position of a star is not fixed but changes due to the varying density and temperature of atmospheric air. This results in the star's position and brightness appearing to change continuously, causing the twinkling effect.
- **Cause:** Changes in the refractive index of air due to temperature variations cause light to bend differently, leading to a slight shift in the star's perceived position.



6.6 Apparent position of a star



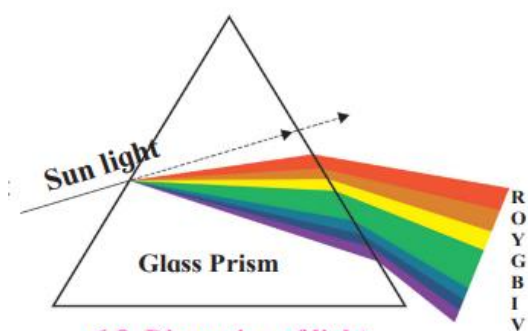
6.7 Effect of atmospheric refraction

Sunrise and Sunset:

- **Phenomenon:**
 - During sunrise and sunset, even when the Sun is below the horizon, its light rays are refracted by the Earth's atmosphere. This bending allows us to see the Sun before it actually rises and after it has set, due to atmospheric refraction (illustrated in Fig 6.7 in the previous image).

Dispersion of Light (Fig 6.8)

- **What is Dispersion?:**
 - Dispersion is the process of splitting white light into its constituent colors when it passes through a medium like a glass prism. The colors appear in the sequence: Violet, Indigo, Blue, Green, Yellow, Orange, and Red (VIBGYOR).



6.8 Dispersion of light

- [Dispersion of light from prism #physics #shorts #dispersion #experiment #science - YouTube](#)

• Key Concepts:

- **Wavelength:**
 - Violet light has the shortest wavelength (~400 nm), while red light has the longest (~700 nm).
- **Refraction:**
 - The speed of light depends on its wavelength when it travels through a medium, causing different colors to bend by different amounts. Violet light bends the most, and red light bends the least.

• Historical Note:

- Sir Isaac Newton demonstrated dispersion by passing sunlight through a glass prism, producing a spectrum of colors.

Partial Reflection:

• Explanation:

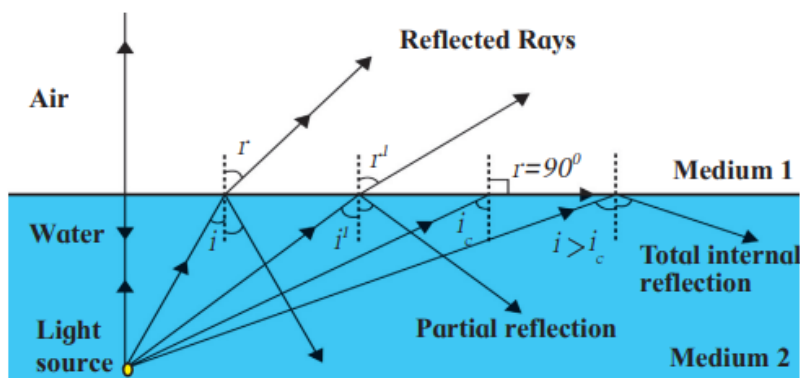
- When light travels from a denser medium (e.g., water) to a rarer medium (e.g., air), part of the light is reflected back into the denser medium, and the rest is refracted into the rarer medium. This phenomenon is known as partial reflection.

Total Internal Reflection:

• Explanation:

- As the angle of incidence increases, the angle of refraction also increases. When the angle of incidence reaches a critical value, the refracted ray travels along the boundary (i.e., the angle of refraction equals 90°). If the angle of incidence is further increased beyond this critical angle, all the light is reflected back into the denser medium. This is known as total internal reflection.

[Total Internal Reflection - YouTube](#)



6.9 Partial and total internal reflection

- Mathematical Expression:**

$${}_1n_2 = \frac{\sin i}{\sin r} \quad \text{For total internal reflection, } i = \text{critical angle, } r = 90^\circ$$

$${}_1n_2 = \frac{\sin i}{\sin 90^\circ} = \sin i \quad (\because \sin 90^\circ = 1)$$

Rainbow Formation (Fig 6.10)

- Process:**

- Rainbows are formed by the combined effects of dispersion, refraction, and total internal reflection of light. When sunlight enters a raindrop, it is refracted and dispersed into its constituent colors. The light undergoes internal reflection within the droplet and is refracted again as it exits. This combination of processes results in the formation of a rainbow.

- Key Concepts:**

- **Dispersion:**
 - The splitting of white light into its component colors.
- **Refraction:**
 - Bending of light as it passes from one medium to another.
- **Total Internal Reflection:**
 - Occurs when the light is completely reflected inside the water droplet, contributing to the intensity and vividness of the rainbow.

